

Lecture 02

Computational Thinking & Algorithms

Chapter lecture — 16 topics with explanations, worked examples, Urdu notes, practice questions and a full answer key.

BIEK Computer Science XII · Bilingual support: English + اردو

بی آئی ای کے کمپیوٹر سائنس — بارہویں جماعت

16 book lectures in this pack

★ Golden topics, Urdu notes, review and answer key

XII

How to use this BIEK study-guide lecture

Each PDF follows one textbook lecture from the BIEK / Sindh Computer Science notes (New Sindh Curriculum). Read *Learn it*, study the SVG diagram, try the worked example on paper, then answer *Check yourself* before the answers. **Extra board practice** (MCQs, short and long questions) is added after the book content — it is not in the textbook. Career-only extras that were not in the book lecture list have been removed. Math is rendered with MathJax. ★ Golden topics are high-yield in board exams. The green box repeats the idea in Urdu.

ہر لیکچر کتاب کے درس کے مطابق ہے۔ کتاب کے بعد اضافی MCQs اور سوالات ہیں۔ جو اضافی کیئریر صفحات کتاب کی فہرست میں نہیں تھے وہ ہٹا دیے گئے ہیں۔

Lectures in this pack

Chapter 2: Computational Thinking & Algorithms. Study one lecture at a time, then use the chapter review and answer key at the end of this PDF.

L01. 2.1 Analyze algorithms for correctness

L03. 2.2 Evaluating the Clarity of an Algorithm

L05. 2.3 Assess Algorithm Efficiency

L07. 2.3.2 Number of Conditions

L09. 2.3.4 The Efficiency Evaluation Framework

L11. 2.5 Concept of Data Structure

L13. 2.5.2 Non-Linear Data Structures (Tree, Graph)

L15. 2.6 Identify data structures in problem scenarios

L02. 2.1.1 – 2.1.3 Trace Tables vs Stepwise Reasoning

L04. 2.2.1 – 2.2.2 Modularity and Readability

L06. 2.3.1 Number of Steps (Big O Notation)

L08. 2.3.3 Number of Repetitions

L10. 2.4 Refinements to improve clarity and efficiency

L12. 2.5.1 Linear Data Structures (Array, Linked List, Stack, Queue)

L14. 2.5.3 Operations on Data Structures

L16. 2.7 Trace data retrieval in Array, List and Queue

Computational Thinking & Algorithms

Class XI taught you how to write algorithms. This chapter teaches you how to **judge** them: is the algorithm correct, clear and efficient? You will then learn the main **data structures** — array, linked list, stack, queue, tree and graph — so you can store data the smart way and trace how it is retrieved.

گیارہویں جماعت میں آپ نے الگورتھم لکھنا سیکھا۔ اس باب میں آپ سیکھیں گے کہ الگورتھم درست، واضح اور تیز ہے یا نہیں، اور ایرے، لنکڈ لسٹ، سٹیک، کیو، ٹری اور گراف جیسے ڈیٹا سٹرکچرز میں ڈیٹا کیسے رکھا اور نکالا جاتا ہے۔

What you will learn

- Define **correctness** and prove it with a complete **trace table**.
- Evaluate **clarity** using modularity (SRP, coupling, abstraction) and readability (names, flow, consistency).
- Explain how the number of **conditions** and **repetitions** changes efficiency, and apply the three-question efficiency framework.
- Define a **data structure** and classify it as linear or non-linear.
- Draw a tree and a graph, and name the five operations on data structures.
- Apply **stepwise reasoning** and compare it with a trace table on at least four points.
- Define **efficiency** in time and space, and state Big O for $O(1)$, $O(\log n)$, $O(n)$, $O(n \log n)$ and $O(n^2)$ with examples.
- Refine an algorithm for clearer names, modules, comments and fewer wasted steps.
- Draw and trace an array, linked list, stack (PUSH/POP) and queue (ENQUEUE/DEQUEUE).
- Choose a data structure for a real scenario and trace retrieval in an array, a linked list and a queue.

2.1 Analyze algorithms for correctness

LEARN IT

An algorithm is **correct** if its **output matches the expected results** for a given input. Analysing correctness means showing that the algorithm produces the intended output for **all valid inputs**, not just one lucky example. Two methods are used in this course:

Trace tables → paper execution, line by line → Stepwise reasoning → logic blocks, overall state

You already met the idea of a dry run in Class XI. Here you learn to do it formally, and to choose the method that fits the size of the algorithm.

WORKED EXAMPLE

Worked example: what "correct" means. An algorithm is supposed to count how many of the heights 1.91, 2.01, 1.87, 1.57, 1.48 lie between 1.57 m and 1.91 m inclusive.

Expected output: 3 (1.91, 1.87 and 1.57 qualify; 2.01 is too tall; 1.48 is too short).

If a program prints 4, it **runs** but it is **not correct**. Correctness is about the right answer, not about "no crash".

REMEMBER IT

Remember: **runs ≠ correct**. A logical error still produces output — just the wrong output. That is why we analyse correctness on paper before we trust the code.

COMMON MISTAKE

Mistake: testing only one "normal" input and calling the algorithm correct.

Fix: also try edge cases (empty list, $N = 0$, the exact boundary values 1.57 and 1.91). The definition says **all valid inputs**.

اردو میں سمجھیں

الگورتھم تب درست (Correct) سمجھا جاتا ہے جب دیے گئے ان پٹ پر اس کا آؤٹ پٹ وہی ہو جو ہونا چاہیے۔ صرف چل جانا کافی نہیں؛ منطقی غلطی سے غلط جواب بھی نکل سکتا ہے۔ درستگی جانچنے کے دو طریقے ہیں: ٹریس ٹیبل (Trace Table) اور مرحلہ وار استدلال (Stepwise Reasoning)۔

CHECK YOURSELF

1. When is an algorithm considered correct?
2. Name the two methods used to analyse correctness in this chapter.
3. Why is "the program ran without an error" not enough?

Answers: 1. When its output matches the expected results for a given input (and does so for all valid inputs). 2. Trace tables and stepwise reasoning. 3. Because a logical error can still produce output — the wrong output.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. An algorithm is correct if:

- a. Outputs match the spec for the intended inputs (incl. edges)
- b. It is short
- c. It uses XOR
- d. It prints Hello

MCQ 2. A failed edge case means:

- a. Not fully correct
- b. Still correct enough always
- c. Faster Big O
- d. Better UX

MCQ 3. You demonstrate correctness in this chapter with:

- a. Trace tables / stepwise reasoning / tests
- b. Only careers
- c. Only Figma
- d. Only pie charts

MCQ 4. A spec is:

- a. The expected behaviour
- b. The CPU model
- c. The font
- d. The sprint name

MCQ 5. Partial correctness often means:

- a. If it halts, the answer is right — you still need termination
- b. It never needs to halt
- c. It is Agile
- d. It is a GUI

MCQ 6. Off-by-one in a loop is a:

- a. Correctness bug
- b. OSI issue
- c. NAND issue
- d. Career issue

Extra short questions

1. Define correctness.
2. Two ways it can fail.
3. Why test edges.
4. Link to later trace tables.

Extra long question

1. Give a "find maximum" algorithm, two traces (normal and $n=1$), and an incorrect version that fails on negatives. Explain.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. For every specified input, the algorithm produces the specified output and finishes.
2. Wrong answer; infinite loop (does not finish).
3. Empty, one element, already sorted, not found, max int.
4. A dry run is a manual proof for a small case.

Long

1. Init max to first element not 0. Trace [3,-1,5] and [-4]. Bug: init 0 fails all-negative lists. That is correctness, not Big O.

2.2 Evaluating the Clarity of an Algorithm

LEARN IT

Clarity means an algorithm is easy to **read, understand and modify**. In many cases, evaluating clarity is even more important than making the algorithm slightly faster. A clear algorithm is easier to **maintain, debug and extend** over time.

The two key aspects of clarity are:

Clarity → Modularity (parts) → Readability (text and flow)

WORKED EXAMPLE

Worked example: speed versus clarity. Algorithm A finds the class average in 8 confusing lines with variables a , b , c . Algorithm B uses two named functions and takes the same number of additions. Both are $O(n)$. Algorithm B is the better school solution because the next person (or you, in three months) can change it without breaking it.

REMEMBER IT

Ask two questions: “Can I split this into parts?” (modularity) and “Can another student read it aloud and understand it?” (readability).

COMMON MISTAKE

Mistake: treating a clever one-line trick as “better” than a clear five-line solution.

Fix: the source says clarity is often more important than speed. Board answers should praise meaningful names and modules, not compacted code.

اردو میں سمجھیں

وضاحت (Clarity) کا مطلب ہے الگورتھم پڑھنا، سمجھنا اور بدلنا آسان ہو۔ بہت سی صورتوں میں وضاحت رفتار سے بھی اہم ہے۔ اس کے دو حصے ہیں: ماڈیولریتی (Modularity) یعنی کام کو حصوں میں بانٹنا، اور پڑھائی پن (Readability) یعنی متن اور پہاڑ صاف ہونا۔

CHECK YOURSELF

1. What does clarity mean?
2. Name the two key aspects of clarity.
3. Why can clarity matter more than a small speed gain?

Answers: 1. The algorithm is easy to read, understand and modify. 2. Modularity and readability. 3. Because a clear algorithm is easier to maintain, debug and extend over time.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Clarity means the algorithm is:

- a. Easy to read, understand, change
- b. As short as possible even if cryptic
- c. Written in one letter variables only
- d. Uncommented hex

MCQ 2. Modularity helps clarity because:

- a. Named pieces have one job
- b. Everything is in main
- c. There are no names
- d. Gotos everywhere

MCQ 3. Readability includes:

- a. Names, structure, comments, consistent style
- b. Removing all spaces
- c. 1-letter names always
- d. No line breaks

MCQ 4. A 200-line monoblock is:

- a. Low modularity
- b. High modularity
- c. A primary key
- d. A pie

MCQ 5. Clarity can matter more than a tiny speed gain when:

- a. Humans must maintain the code
- b. $n=10^{12}$ always
- c. It is a one-off NAND
- d. It is OSI layer 1

MCQ 6. Poor names are a:

- a. Readability problem
- b. TCP problem
- c. Power problem
- d. RAM manufacturing problem

Extra short questions

1. Define clarity of an algorithm.
2. Two ways to improve it.
3. Can a correct algorithm be unclear?
4. Link to functions in Ch 3.

Extra long question

1. Rewrite a messy “average of array” monoblock as three modules (input, compute, output). Explain the clarity gain.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. A person can see the purpose and logic without running it.
2. Decompose into modules; meaningful names; comments on why; consistent structure.
3. Yes — and then it will be modified wrongly.
4. Functions are modularity in Python.

Long

1. Show before/after pseudocode. Each module tested. Names: `sum_list`, `mean`. Comments why $n=0$ is rejected.

2.2.1–2.2.2 Modularity and Readability

LEARN IT

Modularity means breaking an algorithm into smaller, independent parts such as **functions, classes or modules**. This makes the algorithm more organised and easier to work with.

How to evaluate modularity

1. **Single Responsibility Principle (SRP)**. Each module should focus on **one** specific task. If a function calculates a sum, prints a report and saves a file, it violates SRP.
2. **Coupling**. Modules should be reusable in different contexts. Good modularity means **loose coupling**: a change in one part does not smash the others. Tight coupling is when two functions share so many hidden details that you cannot reuse one without the other.
3. **Abstraction**. A well-designed algorithm hides complex implementation details. You can call a sorting function without knowing whether it uses Bubble Sort or Selection Sort inside.

WORKED EXAMPLE

Worked example: average of an array.

Poor modularity (monoblock)

All logic sits in one block, so you cannot reuse the sum.

```
Step 1: Start
Step 2: Set A = [1, 2, 3, 4, 5]
Step 3: Set n = 5
Step 4: Set total = 0
Step 5: Set avg = 0
Step 6: For i = 1 to n
Step 7:   total = total + A[i]
Step 8: End For
Step 9: avg = total / n
Step 10: Print avg
Step 11: End
```

Good modularity (Functions)

Two clear functions: `CalculateSum()` and `FindAverage()`.

```
Step 1: Start
Step 2: Set A = [1, 2, 3, 4, 5]
Step 3: Set n = 5
Step 4: Print FindAverage(A, n)
Step 5: End
Function FindAverage(A, n)
  Sum = CalculateSum(A, n)
  Avg = Sum / n
  Return Avg
End Function
Function CalculateSum(A, n)
  Set Total = 0
  For i = 1 to n
    Total = Total + A[i]
  End For
  Return Total
End Function
```

SRP check: `CalculateSum` only adds; `FindAverage` only divides. Loose coupling: you can reuse `CalculateSum` to find a total without computing an average. Abstraction: the main algorithm just calls `FindAverage` and does not see the loop.

The source pseudocode numbers array positions from 1. Python numbers them from 0, so the loop is `For i in range(n)`.

```
def find_average(A, n):
    total = 0
    for i in range(1, n+1):
        total = total + A[i]
    return total

def find_average(A, n):
    return calculate_sum(A, n) / n

A = [1, 2, 3, 4, 5]
n = 5
print(find_average(A, n))
```

output
2.8

REMEMBER IT

SRP is one line: **one function, one job**. If you cannot name the function with a short verb-phrase (`CalculateSum`, `FindAverage`), it is doing too much.

COMMON MISTAKE

Mistake: copying the 1-based loop `for i = 1 to n: total = total + A[i]` into Python. Then `A[i]` skips the first item and `A[5]` crashes.

Fix: In Python use `for i in range(n)` so the indices are 0, 1, 2, 3, 4.

آردن میں سمجھیں۔

ماتریکس کا مطلب اگرچہ ہم کو آگے حصوں (فکشن کتابیں، مائیکرو) میں پڑتا ہے، مبالغہ کی بنی اصول: واحد قدم داری (SRP) یہی ایک فکشن ایک کام لکھی جات (Extreme Coupling) (تاک ایک حصہ بنانی سے دوسرے) د لوہی اور اجزائے (Abstraction) یہی اندازوں تفصیل دینا کہ صرف کام ہوگا۔۔۔ جسے سوچتے ہوئے بغیر یہ جانی کہ ہیل یہ یا سلیکٹیں۔

CHECK YOURSELF

1. What does SRP say each module should do?
2. Is the good average algorithm, which function hides the loop?
3. Why is the monoblock harder to reuse?

Answers: 1. One specific task. 2. `CalculateSum`. `FindAverage` only divides; the main program only prints it. 3. All logic sits in one block, so you cannot call "just the sum" from another algorithm.

LEARN IT

Readability is how easily another person can understand the purpose and logic of an algorithm **without running it**. It is about how clear the text and the flow are.

How to evaluate readability

1. **Naming conventions** — consistent rules for names. Good names make an algorithm easier to search and understand.

Bad: `v = d / t`

Good: `velocity = distance / time`

2. **Self-documentation** — the algorithm explains itself, so you need fewer comments. Meaningful names and clean formatting make the purpose obvious.

3. **Control-flow simplicity** — keep the logic straightforward. Avoid deep nesting (if inside if inside if), limit extra conditions, and keep each function on one task.

4. **Consistency** — one style throughout: indentation, spacing, and casing (camelCase or snake_case). Then the logic stands out instead of the mess.

Names → Self-doc → Simple flow → Consistent

WORKED EXAMPLE

Worked example: the same formula, two styles.

```
# Read it: need
# 100
# 4
# 25
# v = d / t
# v = 100 / 4
# if v > 10:
#     print(v)
```

output
25.0

```
# Readable
distance = 100
time = 4
velocity = distance / time
speed_limit = 20
if time > 0 and velocity > speed_limit:
    print(velocity)
```

output
25.0

The second version uses names, avoids an if nested inside an if, and keeps one consistent snake_case style. You can understand it without running it. Both print 25.0, but only the second is clear.

REMEMBER IT

Remember readability as **N-D-S-C**: Names, Self-documentation, Simple flow, Consistency.

COMMON MISTAKE

Mistake: dumping a comment on every line (`total = total + 1 # add one to total`).

Fix: If the name already says it, skip the comment. Comments should explain **why**, not repeat **what**.

آردن میں سمجھیں۔

پڑھائی پر یہ کہ دوسرا شخص اگرچہ ہم پہلے بغیر اس کا مشقہ سمجھ لی ابھی تاہم (velocity = distance / time) خود بیان (کہ تجربہ) سادہ کنٹرول فلو (گری کی بغیر) اور ایک جیسا انداز (casing اور indentation) اسے آسان بناتی ہیں۔

CHECK YOURSELF

1. Rewrite `v = d / t` with readable names.
2. What is deep nesting, and why is it a readability problem?
3. Name two style habits that support consistency.

Answers: 1. `velocity = distance / time`. 2. If inside if inside if (or similar); it makes the flow hard to follow. 3. Any two of: consistent indentation, spacing, and casing (camelCase or snake_case).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Modularity is:

- a. Splitting into independent places with clear interfaces
- b. One file of 2000 mixed lines
- c. Random copy-paste
- d. A chart type

MCQ 2. A module should have:

- a. One job
- b. Every job
- c. No name
- d. No inputs

MCQ 3. Readability is:

- a. How easily another person understands the algorithm
- b. Font size of the PDF only
- c. RAM size
- d. Cable length

MCQ 4. Reusing a module is easier if:

- a. It does not depend on hidden globals
- b. It prints inside and uses global v
- c. It is unnamed
- d. It mixes I/O and maths

MCQ 5. Comments should explain:

- a. Why, not every obvious `in=1`
- b. Nothing
- c. Only your WhatsApp
- d. Only Big O symbols

MCQ 6. Poor readability increases:

- a. Bugs when changing code
- b. CPU time
- c. OSI layers
- d. Pie slices

Extra short questions

1. Define modularity and readability.
2. One benefit of each.
3. A readable name vs a bad one.
4. How they interact.

Extra long question

1. Take a messy attendance script (global, 1-letter names). Produce a modular readable version and a 5-point rubric a teacher could mark.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Modularity: parts. Readability: understandable text/structure.
2. Modularity: testhouse. Readability: fewer maintenance bugs.
3. total_marks vs x1.
4. Global named modules are easier to read.

Long

1. Modules: load, mark, save, report. Rubric: names, functions, no magic numbers, comments why, handles empty file.

2.3 Assess Algorithm Efficiency ★ Golden

LEARN IT

Algorithm efficiency is measured in terms of **time** and **space** — how fast it runs and how much memory it uses.

The key considerations for assessing efficiency are:

1. **Number of steps** (described with **Big O notation**).
2. **Number of conditions** (IF decisions).
3. **Number of repetitions** (loops).

Time (speed) + Space (memory) → Efficiency

WORKED EXAMPLE

Worked example: two ways to find a name in 1,000,000 sorted records.

Linear search checks about 1,000,000 names in the worst case (time grows with n). Binary search checks about 20 names (it halves the list each time). Both can be correct and clear. They are not equally efficient.

REMEMBER IT

Efficiency is always “as the input grows”. An algorithm that feels instant on 10 items can freeze on 10 million items if it is $O(n^2)$.

COMMON MISTAKE

Mistake: judging efficiency by how short the code looks.

Fix: a five-line nested loop can do n^2 work. Count steps, conditions and repetitions — not lines of text.

اردو میں سمجھیں

کارکردگی (Efficiency) وقت (رفتار) اور جگہ (میموری) سے ناپی جاتی ہے۔ تین چیزیں دیکھی جاتی ہیں: قدموں کی تعداد (پگ او)، شرطوں کی تعداد، اور دہرائی (لوپ) کی تعداد۔ چھوٹے ان پٹ پر تیز لگنے والا الگورتھم بڑے ان پٹ پر سست پڑ سکتا ہے۔

CHECK YOURSELF

1. Efficiency is measured in terms of which two resources?
2. List the three key considerations named in the source.

Answers: 1. Time (speed) and space (memory). 2. Number of steps (Big O), number of conditions, number of repetitions.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Efficiency is mainly:

- a. Time and extra memory vs n
- b. How pretty the UI is
- c. How many comments
- d. Font choice

MCQ 2. As n grows, we care about:

- a. Growth class (Big O), not a 2 ms blip
- b. Only the constant 2 ms
- c. Only variable names
- d. Only colours

MCQ 3. Space efficiency is:

- a. Extra RAM/disk beyond the input
- b. Screen size
- c. PDF margins
- d. Cable thickness

MCQ 4. An algorithm can be:

- a. Correct but inefficient
- b. Efficient but never correct and still useful always
- c. A pie
- d. A sprint

MCQ 5. This chapter's framework later uses:

- a. Steps, conditions, repetitions
- b. Only UX
- c. Only ER
- d. Only Figma

MCQ 6. Why efficiency is ★:

- a. Board likes compare-two-algorithms questions
- b. It replaces correctness
- c. It replaces HCI
- d. It replaces Python

Extra short questions

1. Define time and space efficiency.
2. Why not measure only one run on your PC?
3. Example of wasting space.
4. Example of wasting time.

Extra long question

1. Compare linear vs binary search on $n=2$, $n=1\,000\,000$ for time and space. State preconditions. 10-mark table.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Time: how running time grows with input. Space: extra memory.
2. Machines differ; we need growth vs n .
3. Copying a huge list on every recursive call.
4. Linear search 1 million times on a static sorted list instead of binary/index.

Long

1. Time $\sim n$ vs $\sim \log n$. Space $O(1)$ both (iterative). Binary needs sorted. For $n=1e6$, $\sim 1e6$ vs ~ 20 comparisons. Correctness not sacrificed.

2.3.1 Number of Steps (Big O Notation) 🔗

LEARN IT

When we discuss efficiency we ask how performance changes as the **input size** n increases. **Big O notation** is a mathematical method used to describe how the running time (or memory) **grows** as n grows. We ignore small constants and keep the most important term. If an algorithm does $5n + 5$ steps we write $O(n)$. In computer science, **log** n almost always means **log base 2** (how many times you can halve n until you reach 1).

Five common classes

Class	Name	What happens when n grows	Typical example	Grade
$O(1)$	Constant	Time stays the same for 10 items or 10 million.	Access <code>arr[5]</code> (or any known index).	Excellent
$O(\log n)$	Logarithmic	Time grows very slowly; the problem is halved each step.	Binary search on a sorted list.	Great
$O(n)$	Linear	If n doubles, time roughly doubles.	Linear search of an unsorted list.	Fair / Good
$O(n \log n)$	Linearithmic	A little worse than linear; much better than n^2 . Common in good sorting.	Merge Sort (split in half, then merge).	Good
$O(n^2)$	Quadratic	Time grows with the square of n . Nested loops. Painful for large n .	Bubble Sort; compare every item with every other item.	Poor

The source stresses $O(1)$, $O(\log n)$, $O(n)$ and $O(n \log n)$ is the class of efficient general-purpose sorts, so you should recognise it when a question says "better than Bubble Sort but more than linear".

Growth table (log is $\log_2$)

Each cell is the approximate number of operations for that n . Read down a column to see how fast that class explodes.

n	$O(1)$	$O(\log n)$	$O(n)$	$O(n \log n)$	$O(n^2)$
2	1	1	2	2	4
4	1	2	4	8	16
8	1	3	8	24	64
16	1	4	16	64	256
32	1	5	32	160	1,024
1,024	1	10	1,024	10,240	1,048,576
1,000,000	1	~20	1,000,000	~20,000,000	1,000,000,000,000

From $n = 8$ to $n = 1,000,000$, $O(1)$ does not move, $O(\log n)$ only goes from 3 to about 20, $O(n)$ becomes a million, and $O(n^2)$ becomes a million million — which is why nested loops "fail for large datasets".



Same $n = 16$. Quadratic work is already four times $n \log n$ and sixteen times linear.

WORKED EXAMPLE

Worked example A — $O(1)$: first element of an array. Whether the array holds 10 or 10 million values, `arr[5]` is one jump to a known address.

```
arr = [21, 76, 12, 99, 24, 48]
print(arr[5])
```

array
21

Worked example B — $O(n)$: linear search. Each extra item adds one extra check.

```
def linear_search(items, key):
    for i, value in enumerate(items):
        if value == key:
            return i
    return -1

print(linear_search([11, 50, 20, 76, 96, 86, 201, 201])
print(linear_search([19, 50, 20, 76, 96, 201, 99])
```

array
-1

Worked example C — $O(\log n)$: binary search on a sorted list. Each step throws away half of the remaining items. For 8 sorted numbers the worst case is about 4 comparisons (from $\log_2 8 = 3$).

```
def binary_search(arr, key):
    low = 0
    high = len(arr) - 1
    while low < high:
        mid = (low + high) // 2
        if arr[mid] == key:
            return mid
        elif arr[mid] < key:
            low = mid + 1
        else:
            high = mid - 1
    return -1, steps

data = [1, 6, 12, 17, 22, 31, 48, 51]
print(binary_search(data, 40))
print(binary_search(data, 33))
```

array
(4, 3)

array
(5, 3)

Key 40 is found at index 6 after 3 comparisons. Key 3 is found at index 0 after 3 comparisons. Never more than a handful of steps, even if the list later grows to a million (about 20 steps).

Worked example D — $O(n \log n)$: Merge Sort idea. To sort 8 items, Merge Sort makes $\log_2 8 = 3$ split-levels, and each level walks all 8 items once, so work = $8 \times 3 = 24$, which matches the growth table. Bubble Sort on 8 items is on the order of $8^2 = 64$ comparisons. That gap becomes enormous at $n = 1,000,000$.

Worked example E — $O(n^2)$: nested loops.

```
n = 6
count = 0
for i in range(n):
    for j in range(i):
        count += 1
print(count)
```

count
15

Inner body runs $n \times n = 36$ times. That is the pattern of Bubble Sort and of "compare every item with every other item".

REMEMBER IT

Order from fastest to slowest (for large n): $O(1) < O(\log n) < O(n) < O(n \log n) < O(n^2)$

Mnemonic: "Constant Logs Never Need Squares" — Constant, Logarithmic, N , $N \log N$, Squares.

COMMON MISTAKE

Mistake 1: writing $O(n)$ for binary search because "there is a loop". The loop runs about $\log n$ times, not n times.

Mistake 2: writing $O(n)$ or $O(n \log n)$. Big O drops constants both are $O(n)$.

Mistake 3: using binary search on an unsorted list. Binary search is $O(\log n)$ only on sorted data.

آردو میں سمجھیں

گت او (Big O) بتاتا ہے کہ ان قیمت ۵ پر وقت یا کمپیوٹر کیسے بڑھیں گی۔ $O(1)$ مستقل (ایر-ٹو) کا $O(\log n)$ پر قدم پر آگے (ڈائری سرچ) $O(n)$ کیلکولی (ڈائری سرچ) $O(n \log n)$ بعض حالت میں سرچ کرتا ہے اور $O(n^2)$ مربع (ہیل مارٹل اندرونی ٹورس) ہے۔ n پر $O(n^2)$ تقابلی استعمال ہو سکتا ہے۔

EXAM TIP

Memorise the five rows of the class table (name, growth, one example, grade). Typical MCQ "Accessing `arr[0]` is ...?" = $O(1)$. "Binary search is ...?" = $O(\log n)$. "Nested loops / Bubble Sort ...?" = $O(n^2)$. Long questions ask you to fill or interpret a growth table, or to say what happens if n becomes 10 times larger (linear = 10x time; quadratic = 100x time). Always mention that $\log n$ is "halving" and n^2 is "nested loops".

CHECK YOURSELF

1. Give one example each of $O(1)$, $O(\log n)$, $O(n)$ and $O(n^2)$.

2. From the growth table, how many operations does $O(n^2)$ need when $n = 32$?

3. Why is Merge Sort described as $O(n \log n)$?

Answers: 1. $O(1)$ = `arr[0]`, `len(arr)`; binary search; $O(\log n)$ linear search; $O(n)$ Bubble Sort / nested loops; 2. 1,024; 3. It makes about $\log n$ split-levels and does about n work on each level, so $n \times \log n$.

Extra board practice These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key. MCQ 1. $O(n)$ means: a. Constant time, independent of n b. Linear in n c. Quadratic d. Factorial	MCQ 2. $O(n)$ example: a. Linear search b. Binary search c. Needs $n \times n$ always d. Even $O(1)$ need always guaranteed	MCQ 3. $O(n \log n)$ example: a. Binary search b. Bubble sort c. Reading all n items once d. Nested loops n^2	MCQ 4. $O(n^2)$ example: a. Bubble / selection sort b. Binary search c. Closed formula sum d. Array index	MCQ 5. We drop constants and small terms because: a. Growth class dominates for large n b. They are always 0 c. Big O is UK d. Bounds forbid numbers	MCQ 6. $\log n$ grows: a. Slowly b. Faster than n^2 c. Faster than n d. As a straight 45° line always
--	---	---	---	---	---

Extra short questions

1. Define Big O in one student sentence.
2. Order these from fast to slow: n^2 , 1, $\log n$, n .
3. Why is $3n+5$ still $O(n)$?
4. What does $O(n \log n)$ look like on a phone contacts list?

Extra long question

1. For each array index, binary search, linear search, bubble sort — state Big O and a 1-line why. Then pick one for $n=50$ vs $n=5$ million. ★ Golden.

Extra answer key MCQs: 1. A, 2. A, 3. A, 4. A, 5. A, 6. A Short 1. As n increases, n^2 (or time for space) grows as n gets large. 2. $O(1)$, $O(\log n)$, $O(n)$, $O(n^2)$ 3. Constants ignored; linear term dominates. 4. Each step halves the remaining names. Long 1. $O(1)$, $O(\log n)$, $O(n)$, $O(n^2)$ = 50 anything fine, $n=50$ avoid n^2 , prefer $\log$ /linear as appropriate. Mention sort cost if you need binary.
--

2.3.2 Number of Conditions

LEARN IT

Conditions (IF statements) are decision points. Every condition forces the CPU to evaluate True or False, so excessive branching reduces performance.

- **Computational cost.** Each IF is extra work, even if the rest of the algorithm is simple.
- **Nested conditions.** IF inside IF often means a brute-force approach. It hurts both readability and speed.
- **Short-circuiting** (the optimisation goal). Stop evaluating as soon as the result is known. In a logical AND, if the first part is False the second part is skipped. In a logical OR, if the first part is True the second part is skipped.

WORKED EXAMPLE

Worked example: cheap test first. You only divide if the denominator is not zero. Place the cheap, safe test on the left so Python never evaluates the dangerous part.

```
x = 0
y = 10
if x != 0 and (y / x) > 2:
    print("ratio is 'large'")
else:
    print("skipped safely")
```

OUTPUT
skipped safely

$x \neq 0$ is False, so AND short-circuits and y / x is never attempted. If you wrote the division first, the program would crash. Same idea in the soldier algorithm: a False height test skips Step 6.

REMEMBER IT

Put the **cheapest / most likely False** test first in an AND, and the **most likely True** test first in an OR. That is short-circuiting used on purpose.

COMMON MISTAKE

Mistake: writing a chain of nested IFs when one compound condition would do (as in the velocity example in 2.2.2).

Fix: flatten with AND/OR, and rely on short-circuiting instead of extra layers.

آردو میں سمجھیں

IF پر پروسیسر سے True/False پوچھتی ہے اس لیے زیادہ شاخیں کارکردگی گھٹاتی ہیں۔ اندرونی IF اکثر brute-force ہوتی ہے۔ ہارٹ سرکٹ (Short-circuit) کا مطلب ہے جیسے ہی جواب معلوم ہو جانچ روک دی جائے، مثلاً AND میں پہلی شرط False ہو تو دوسری نہ چلے۔

CHECK YOURSELF

1. Why do extra IF statements reduce performance?
2. What does nested IF often indicate?
3. In $x \neq 0$ and $(y / x) > 2$ with $x = 0$, why is there no division-by-zero?

Answers: 1. Each condition must be evaluated True/False by the CPU. 2. A brute-force approach (and it reduces readability and speed). 3. AND short-circuits: the first part is False, so the division is never evaluated.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Number of conditions affects:

- a. How many decisions (and often branches) you pay for
- b. Only the font
- c. Only RAM manufacturing
- d. Only UX colour

MCQ 2. A nested if inside a loop can make work:

- a. Depend on data (sometimes skip inner work)
- b. Always n^2
- c. Always $O(1)$
- d. Always crash

MCQ 3. Short-circuit and/or can:

- a. Skip later tests
- b. Sort the array
- c. Open a file
- d. Draw a pie

MCQ 4. Too many rare special cases:

- a. Hurt clarity and can hurt time if mis-ordered
- b. Always reduce Big O class
- c. Always increase RAM
- d. Always fix UX

MCQ 5. Binary search's conditions:

- a. Few, but they cut n in half — very efficient
- b. n^2
- c. Are UX heuristics
- d. Are SQL joins

MCQ 6. Counting conditions is part of the:

- a. Efficiency evaluation framework
- b. HCI Nielsen list only
- c. ER cardinality only
- d. MVP canvas only

Extra short questions

1. What is a "condition" in this chapter's sense?
2. How can conditions save time?
3. How can they waste time?
4. Link to readability.

Extra long question

1. Two versions of search: (A) no break (B) break when found. Compare conditions, time, traces on a hit at position 2 of 10.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. A true/false test the algorithm performs (if, loop test, while).
2. Early exit: if found, break; if $x < 0$, skip expensive work.
3. Re-testing the same expensive predicate; poorly ordered or/and.
4. A tower of nested ifs is slow to understand even if $O(n)$.

Long

1. A always 10 tests. B 2 tests. Same Big O class worst-case still $O(n)$, but average better. Show trace tables.

2.3.3 Number of Repetitions

LEARN IT

Repetitions (loops) are the most significant factor in algorithm efficiency. The **structure of the loop** decides how time grows with n .

Type of repetition	Conceptual impact	Efficiency grade
Linear loop	Processes each element exactly once. Time grows in proportion to size.	Good — $O(n)$
Logarithmic repetition	Reduces the problem size by half each iteration (binary search). Huge time savings.	Excellent — $O(\log n)$
Nested loops	For each element, process all other elements. Operations explode.	Poor — $O(n^2)$

WORKED EXAMPLE

Worked example: count the inner-body runs.

```
n = 5
linear = 0
for i in range(n):
    linear = linear + 1

nested = 0
for i in range(n):
    for j in range(n):
        nested = nested + 1

print(linear, nested)
```

```
OUTPUT
5 25
```

One loop: 5. Nested loops: $25 = 5^2$. For $n = 1,000$ that would be 1,000 versus 1,000,000.

REMEMBER IT

Find the **deepest loop**. One loop $\approx O(n)$. Loop inside a loop $\approx O(n^2)$. A loop that halves $n \approx O(\log n)$.

COMMON MISTAKE

Mistake: counting the word `for` and ignoring how the index moves.

Fix: a loop `while n > 1: n = n // 2` is logarithmic, even though it is still "a loop".

اردو میں سمجھیں

دہرائی (لوپ) کارکردگی کا سب سے بڑا عامل ہے۔ ایک بار پر عنصر پر چلنے والا لوپ $O(n)$ (اچھا)، ہر بار مسئلہ آدھا کرنے والا $O(\log n)$ (بہترین)، اور اندرونی لوپ $O(n^2)$ (خراب) ہے۔ اندرونی جسم کے چلنے کی گنتی سے یہ فرق صاف نظر آتا ہے۔

CHECK YOURSELF

- What grade does the source give nested loops?
- Why is binary search's loop logarithmic rather than linear?
- If $n = 5$, how many times does a fully nested n -by- n pair of loops run?

Answers: 1. Poor — $O(n^2)$. 2. Each iteration halves the remaining problem, so the number of iterations is about $\log n$. 3. 25.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. for i in range(3) runs:

- $i=0,1,2$
- $i=1,2,3$
- $i=3,2,1$
- forever

MCQ 2. while repeats while the condition is:

- True
- False
- A file
- None always

MCQ 3. break:

- Exits the loop immediately
- Skips one line only
- Closes Python
- Commits SQL

MCQ 4. continue:

- Skips the rest of this iteration
- Ends the program
- Defines a function
- Opens a file

MCQ 5. A nested loop: inner runs:

- Fully for each outer step
- Once in total
- Never
- Only if NAND

MCQ 6. range(2, 6) yields:

- 2,3,4,5
- 2,3,4,5,6
- 1,2,3,4,5,6
- 6,5,4

Extra short questions

- When for vs while?
- What is an infinite loop?
- Trace $s=0$; for i in range(1,4): $s+=i$.
- Shopping bill uses which loop typically?

Extra long question

- Write nested loops to print a 3×3 table of i,j . Then rewrite the shopping-bill outline with a while, a running total, and an if on quantity.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

- for: known count / iterate a sequence. while: unknown count, stop on a condition.
- while True without break, or a condition that never becomes False.
- $i=1$ $s=1$; $i=2$ $s=3$; $i=3$ $s=6$.
- while more items (unknown count) or for a known list of items.

Long

- for i in range(3): for j in range(3): print(i,j). Bill: total=0; while True: read item/qty; if qty<0: continue; if item=="end": break; total+=...; print total.

2.3.4 The Efficiency Evaluation Framework

LEARN IT

To evaluate an algorithm at a conceptual level, ask three questions:

- Does it scale?** If n grows 10 times, does time jump 10 times (linear) or 100 times (quadratic)? Quadratic scaling fails for large datasets.
- What is the bottleneck?** Focus on the most time-consuming section (usually the deepest loop) and try to improve that part.
- Is the cost worth it?** A theoretically faster $O(\log n)$ algorithm can be too complex to code and maintain. For a list of just 50 items, a simple $O(n)$ linear search is practically better because it is simple.

Does it scale?

→

Where is the bottleneck?

→

Is the extra complexity worth it?

WORKED EXAMPLE

Worked example: 50 names versus 1 million names.

Situation	Scale	Bottleneck	Worth the cost?	Choice
50 unsorted class names	Tiny n	A single linear scan	Binary search needs a sort first, which is extra work and extra code	Linear search $O(n)$
1,000,000 sorted account numbers, searched all day	$n \times 10$ compared with 100,000 is already huge; quadratic would be impossible	The search loop	Yes — binary search is standard and well understood	Binary search $O(\log n)$

REMEMBER IT

Framework mnemonic: **S-B-W** — Scale, Bottleneck, Worth it.

COMMON MISTAKE

Mistake: always choosing the theoretically fastest Big O.

Fix: question 3 exists for a reason. For $n = 50$, simple $O(n)$ beats a fragile $O(\log n)$ implementation.

اردو میں سمجھیں

کارکردگی کے تین سوال: کیا الگورتھم بڑے n پر چل سکتا ہے (Scale)؟ سب سے سست جگہ کون سی ہے (Bottleneck)، عموماً گہرا لوپ؟ کیا تیز تر طریقے کی پیچیدگی اس چھوٹے فائدے کے لائق ہے؟ پچاس ناموں کے لیے سادہ $O(n)$ تلاش عملی طور پر بہتر ہو سکتی ہے۔

CHECK YOURSELF

- If n grows $10\times$, how much slower is a quadratic algorithm?
- Where do you usually look for the bottleneck?
- Why might linear search beat binary search for 50 items?

Answers: 1. About $100\times$ slower (10^2). 2. The most time-consuming section, usually the deepest loop. 3. Binary search is more complex to code and maintain, and for $n = 50$ the simplicity of $O(n)$ is practically better.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. The framework looks at:

a. Steps, conditions, repetitions (then Big O)

b. Only logos

c. Only careers

d. Only Figma

MCQ 2. "Number of steps" is:

a. Basic operations in one pass / closed form

b. The page count of the PDF

c. OSI layers

d. Sprint days

MCQ 3. You combine the three to:

a. Justify a Big O class

b. Pick a colour

c. Pick a font

d. Pick a career

MCQ 4. A 5-mark compare question wants:

a. The framework on both algorithms + a verdict

b. Only names

c. Only Python libraries

d. Only UX

MCQ 5. Constants matter at tiny n but the framework emphasises:

a. Growth

b. Hex colours

c. Teacher names

d. PDF margins

MCQ 6. If repetitions are nested $n\times n$, Big O is likely:

a. n^2 regardless of a few extra ifs

b. $\log n$

c. 1

d. $n!$

Extra short questions

- State the efficiency evaluation framework.
- Apply in one line to binary search.
- Apply to bubble sort.
- When would you still pick the slower class?

Extra long question

- Using the framework, compare selection sort vs binary search as if a student mixed them up. Then say why they are not alternatives.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

- Count significant steps, conditions, repetitions; combine into time/space class; compare to requirements.
- Few steps per shot, one/two conditions, $\log n$ repetitions $\rightarrow O(\log n)$ time, $O(1)$ space.
- Compare+swap steps, a condition, nested repetitions $\rightarrow O(n^2)$.
- Tiny n , need stability/simplicity, or the fast one needs a costly precondition (sort first).

Long

- Different problems (order vs find). Framework still: both have loops; sort n^2 vs search $\log n$. Verdict: sort then many binary queries if the list is reused.

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2.4 Refinements to improve clarity and efficiency

LEARN IT

Improving an algorithm means reviewing it step by step: simplifying logic, removing redundant work, and picking smart data structures. A clear algorithm is easier to follow; an efficient one wastes fewer steps.

2.4.1 Refinements for clarity

- 1. Use meaningful names. Replace vague variables with descriptive ones. Bad: $a = b / c$. Good: $average = totalSum / totalItems$.
- 2. Break into modules. Divide a massive block into functions, for example separating `CalculateSum()` from the printing logic.
- 3. Improve structure and indentation. Packed loops and IF statements confuse the reader. Spacing reduces mental effort.
- 4. Add comments. Keep them short. Explain why something is done, not just what is happening, especially for tricky logic.

2.4.2 Refinements for efficiency

- 1. Reduce unnecessary steps. In Bubble Sort, after the first pass the largest value has settled at the end. Ignore that sorted tail on the next pass by shrinking the inner loop, and you save comparisons.
- 2. Use better algorithms. Searching 1 million sorted records with Binary Search instead of Linear Search changes the time from seconds to milliseconds.
- 3. Avoid re-computing values (memoization / caching). If a value never changes, compute it once outside the loop, store it, and reuse it.
- 4. Use appropriate data structures. Direct access → array. Hierarchical data → tree. Do not force one structure everywhere.

WORKED EXAMPLE

Worked example 1 — shrinking Bubble Sort's inner loop ($n = 5$). A naive inner loop always runs $n-1 = 4$ times, for 4 passes: 16 comparisons. The refined inner loop runs 4, then 3, then 2, then 1: 10 comparisons.

```
n = 5
naive = 0
for i in range(n - 1):
    for j in range(n - 1):
        naive = naive + 1

refined = 0
for i in range(n - 1):
    for j in range(n - 1 - i):
        refined = refined + 1

print(naive, refined)
```

```
OUTPUT
16 10
```

Worked example 2 — do not re-compute a constant inside the loop.

```
A = 3
B = 4
N = 5

# BAD: C is recalculated every time
for i in range(1, N + 1):
    C = A + B
    print(C * i)
```

```
OUTPUT
7
14
21
28
35
```

```
A = 3
B = 4
N = 5
# GOOD: compute C once
C = A + B
for i in range(1, N + 1):
    print(C * i)
```

```
OUTPUT
7
14
21
28
35
```

Same output, fewer additions. That is caching: compute once, reuse N times.

REMEMBER IT

Clarity: **N-M-I-C** (Names, Modules, Indentation, Comments). Efficiency: **S-A-C-D** (Shrink steps, Algorithm choice, Cache, Data structure).

COMMON MISTAKE

Mistake: "optimising" by renaming variables and thinking that is an efficiency refinement.

Fix: meaningful names improve clarity. Efficiency refinements change the amount of work (fewer loop iterations, a better algorithm, no re-compute, a better structure).

آرڈو میں سمجھیں

وضاحت بہتر کرنے کے لیے معنی خیز نام، مائیکرو، درست indentation اور مختصر تبصرے استعمال کریں۔ کارکردگی کے لیے فضول قدم ہٹائیں (بیل سارٹ میں ترتیب شدہ دم نظر انداز کریں)، بہتر الگورتھم چنیں (آرڈینری کی جگہ ہائری سرج)، لوپ کے اندر وہی حساب بار بار نہ کریں (Caching/Memoization)، اور مناسب ڈیٹا سٹرکچر رکھیں۔

CHECK YOURSELF

- 1. Rewrite $a = b / c$ with meaningful names for an average.
- 2. After Bubble Sort's first pass, what should the next pass ignore?
- 3. In the A+B example, where should $C = A + B$ be written?

Answers: 1. $average = totalSum / totalItems$. 2. The already-sorted largest value at the end (shrink the inner loop). 3. Once, outside the loop, then reuse C inside.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A clarity refinement is:

- a. Rename, split, comment, simplify logic
- b. Add more nested loops for fun
- c. Shorten names to a,b,c
- d. Remove all spaces

MCQ 2. An efficiency refinement is:

- a. Cut extra work (e.g. already-sorted inner of bubble)
- b. Add random sleeps
- c. Copy the list 12 times
- d. Print inside the inner loop 1e6 times

MCQ 3. Flagged bubble sort's early exit is:

- a. Efficiency refinement
- b. A new UX colour
- c. A foreign key
- d. A pie

MCQ 4. Extracting a function is:

- a. Clarity (and sometimes reuse) refinement
- b. Always worse Big O
- c. OSI layering
- d. A histogram

MCQ 5. Do not "optimise" first if:

- a. The algorithm is still incorrect or unreadable
- b. $n = 10^{12}$ already proven
- c. Tests all pass and profiler points here
- d. It is $O(n!)$ on huge n and you must ship

MCQ 6. Redundant work means:

- a. Computing the same thing twice
- b. A needed loop
- c. A primary key
- d. A caption

Extra short questions

- 1. Two clarity refinements.
- 2. Two efficiency refinements from the book.
- 3. A refinement that hurts clarity for 1% speed is:
- 4. How do you know a refinement worked?

Extra long question

- 1. Show bubble sort before/after: (1) meaningful names (2) last-i unsorted bound (3) swapped flag. State what each changed (clarity vs efficiency).

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

- 1. Better names; extract module; remove duplicate logic; add pre/post comments.
- 2. Stop inner bubble as the suffix grows sorted; avoid extra copies; fewer conditions in the inner loop.
- 3. Usually a bad deal in XU/XII and in teams.
- 4. Same tests (correctness) + fewer operations or clearer reading.

Long

- 1. Code/pseudocode both versions. (1) clarity (2) fewer comparisons $n^2/2$ still $O(n^2)$ (3) best case $O(n)$. Tests unchanged.

2.5 Concept of Data Structure ★ Golden

LEARN IT

A **data structure** is a systematic way of **organising, storing and managing** data so that an algorithm can access, modify and process it efficiently.

Real-life analogy. Books on a shelf are arranged in a set order, so you can find a title quickly. Files in folders are grouped on purpose. Data structures do the same job inside computer memory.

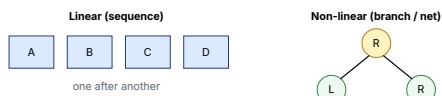
1. Linear data structures

Elements stand in a strict sequence, one after another. Each element (except the first and last) has exactly **one predecessor** and **one successor**.

Examples: Array, Linked List, Stack, Queue.

2. Non-linear data structures

Elements are connected in a hierarchy or a network. One element can connect to **several** others at the same time. Examples: Tree, Graph.



Linear: a line of neighbours. Non-linear: a node with more than one connection.

WORKED EXAMPLE

Worked example: classify each everyday picture.

1. Students standing in a single assembly line → **linear** (each has one in front and one behind).
2. A family tree of grandparents, parents and children → **non-linear** (one parent, several children).
3. Roads linking Karachi, Hyderabad, Sukkur and Larkana in many ways → **non-linear** (a network).

REMEMBER IT

Ask: "Does each item have at most one next item?" Yes → linear. "Can one item link to many, or can links form a net?" → non-linear.

COMMON MISTAKE

Mistake: calling a tree "linear" because you can walk it from top to bottom.

Fix: walking is traversal. The **shape** is hierarchical: one parent, many children. That is non-linear.

اردو میں سمجھیں

ڈیٹا سٹرکچر ڈیٹا کو منظم طریقے سے رکھنے کا نظام ہے تاکہ الگورتھم اسے جلدی استعمال کر سکے۔ لکیری ڈھانچوں میں عناصر قطار میں ہوتے ہیں (ایرے، لنکڈ لسٹ، سٹیک، کیو)۔ غیر لکیری میں ایک عنصر کئی سے جڑ سکتا ہے (ٹری، گراف)۔ الماری کی کتابیں اور فولڈرز کی مثال یاد رکھیں۔

EXAM TIP

Define data structure in full (organise, store, manage; efficient access and processing). Then split linear vs non-linear with the "one predecessor and one successor" test, and list the four linear and two non-linear examples. The shelf / folder analogy earns a mark. MCQs: "Which is non-linear?" → Tree or Graph.

CHECK YOURSELF

1. Define data structure.
2. Name four linear and two non-linear examples.
3. What is true of each middle element in a linear structure?

Answers: 1. A systematic way of organising, storing and managing data for efficient access, modification and processing. 2. Linear: array, linked list, stack, queue. Non-linear: tree, graph. 3. It has exactly one predecessor and one successor.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A data structure is:

- a. An organised way to store and access data
- b. A random pile
- c. A font
- d. A sprint

MCQ 2. We pick a structure for:

- a. The operations we need (fast insert? fast search?)
- b. The logo
- c. The teacher's birthday
- d. The PDF theme

MCQ 3. Linear vs non-linear is about:

- a. Whether elements form a sequence vs branches/links
- b. Colour
- c. File format .py vs .txt
- d. Agile vs waterfall

MCQ 4. An array gives fast:

- a. Index access $O(1)$
- b. Insert at front $O(1)$ always
- c. Arbitrary graph neighbours
- d. Sorted insert $O(1)$ always

MCQ 5. This is ★ because papers ask:

- a. Define + why we need DS + one example
- b. Only careers
- c. Only Figma
- d. Only OSI

MCQ 6. A queue of print jobs is a DS chosen because:

- a. Order of arrival matters (FIFO)
- b. LIFO undo
- c. Hierarchy of boss-staff
- d. Random friends links

Extra short questions

1. Define data structure.
2. Why not always use one big list?
3. Give one linear and one non-linear example.
4. What is an operation on a DS?

Extra long question

1. "If you pick the wrong structure, the algorithm looks harder." Explain with college examples of array, stack, queue, tree. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. A systematic way to organise data for efficient access and modification.
2. Some problems need LIFO, FIFO, hierarchy, or many-to-many links.
3. Linear: array/stack/queue. Non-linear: tree/graph.
4. Insert, delete, search, traverse, peek...

Long

1. Roll list: array. Undo typing: stack. Printer/canteen line: queue. Org chart/folders: tree. Wrong pick: using a stack for a fair queue angers people.

$A(x) = 90$ (index starts at 0)

Step	Operation	Stack before 1st op	Stack after 1st op	Output
0	init()		0	init
1	Push Stack		Stack	Stack
2	Pop() Stack		Stack, Stack	Stack
3	Push Stack		Stack, Stack, Stack	Stack
4	Pop()		Stack, Stack, Stack	Stack

Year	Quarter	Current Order & Back	Lead	Rate	Revenue
0	January	0	—	—	—
1	February	0	0	0	0
2	March	0	0	0	0
3	April	0	0	0	0

2.5.2 Non-Linear Data Structures (Tree, Graph)

LEARN IT

A **tree** organises data in a **hierarchy**. Everything branches out from a single starting point called the **root**.

It creates a strict **one-to-many (parent-child)** relationship. That is an excellent way to show how roles flow in an organisation.

Real-life examples: a family tree (grandparents → parents → children) and folders inside folders on a drive.



College tree: one root (Principal), then parent-child branches. No child has two parents in this model.

WORKED EXAMPLE

Worked example: name the relationships.

- Root: Principal.
- Children of the root: Administrative Head, Academic Dean.
- Children of Academic Dean: HOD Math, HOD CS.
- HOD CS has one parent (Academic Dean), not two — that is the tree rule. If HOD CS also reported sideways to Accounts as an equal link, the picture would become a **graph**.

REMEMBER IT

Tree = family / folders / organogram. One root. Branches down. Parent-child, not a web of equals.

COMMON MISTAKE

Mistake: drawing a tree with two roots, or with an arrow going back up to the Principal from HOD CS (a cycle).

Fix: one root, links only parent → child, no loops. Cycles belong to graphs.

آرڈو میں سمجھیں

ٹری ٹیپا کو درجہ بندی (Hierarchy) میں رکھتی ہے جو ایک (Root) سے پھولتی ہے، تعلق والدین-بچہ (ایک سے کئی) ہوتا ہے، جیسے پرنسپل → ڈین → ایچ او ڈی، خاندانی شجرہ، یا فولڈر کے اندر فولڈر۔ دو چیزیں یا چکر لڑی کو گراف بنا دیتے ہیں۔

CHECK YOURSELF

1. What is the starting node of a tree called?
2. What kind of relationship does a tree model?
3. Give two real-life tree examples from the source.

Answers: 1. The root. 2. One-to-many parent-child. 3. A family tree, and folders inside folders (or a college organogram).

LEARN IT

A **graph** organises data when things connect in a flexible, complex way. Unlike a tree, there is **no strict hierarchy**: any node can connect to any other node.

- **Nodes (vertices)** — the items (cities, people).
- **Edges** — the connections (roads, friendships).

Examples: Facebook friends; GPS mapping (the fastest route between two cities).



Four vertices A, B, C, D. Edges can form a cycle (A-B-D-C-A). A tree cannot do that.

WORKED EXAMPLE

Worked example: Karachi – Hyderabad – Sukkur – Larkana. Cities are vertices. Roads are edges. You can travel Karachi → Hyderabad → Sukkur or Karachi → Larkana → Sukkur. There is no single “root city”. That is a graph, and GPS uses it to pick a fast route.

REMEMBER IT

Tree = Graph in shape-talk: a tree is a special graph with one root and no cycles. If you need cycles or many-to-many links, say **graph**.

COMMON MISTAKE

Mistake: using a tree for GPS because “Hyderabad comes under Sindh”.

Fix: administration can be a tree; **roads** are a graph because you can loop and take alternative paths.

آرڈو میں سمجھیں

گراف لچکدار روابط کے لیے ہے: کوئی سخت درجہ بندی نہیں، کوئی بھی نوڈ کسی سے جڑ سکتا ہے، نوڈز (Vertices) چڑیاں ہیں، کنارے (Edges) تعلق، قریبی یک دوست اور GPS نقشہ، کلاسیکی مثالیں ہیں، چکر (Cycle) اور کئی راستے، گراف کو لڑی سے الگ کرتے ہیں۔

CHECK YOURSELF

1. What are the two parts of a graph?
2. Give two source examples of graphs.
3. State one difference between a tree and a graph.

Answers: 1. Nodes/vertices and edges. 2. Social networks (Facebook friends) and GPS mapping. 3. A tree has a hierarchy and one root; a graph has no strict hierarchy and any node may connect to any other (cycles allowed).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. An array stores items:

- a. In contiguous indexed slots of one type
- b. With arbitrary node links only
- c. As a hierarchy always
- d. As a pie

MCQ 2. A linked list stores items:

- a. In nodes with pointers to the next
- b. In one block of equal indexes only
- c. As a tree of bosses
- d. As a hash of pies

MCQ 3. A stack is:

- a. LIFO
- b. FIFO
- c. A graph
- d. A GUI

MCQ 4. A queue is:

- a. FIFO
- b. LIFO
- c. A binary tree
- d. A K-map

MCQ 5. Undo in an editor uses a:

- a. Stack
- b. Queue
- c. Pie
- d. OSI layer

MCQ 6. A printer line is a:

- a. Queue
- b. Stack
- c. AVL tree always
- d. NAND

Extra short questions

1. Define array, linked list, stack, queue in one line each.
2. One op each: push/pop, enqueue/dequeue.
3. Array vs list for insert at front.
4. Peek means:

Extra long question

1. Draw stack and queue after the sequence: insert A,B,C then remove one. Show remaining order and a real example of each. ★ Golden for stack/queue.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Array: Indexed block. List: nodes+links. Stack: LIFO. Queue: FIFO.
2. Stack push/pop top. Queue enqueue rear, dequeue front.
3. Array shifts $O(n)$; list pointer change $O(1)$ if you have the head.
4. Read the top/front without removing.

Long

1. Stack remaining B,A (C popped). Queue remaining B,C (A served). Examples: back button vs ticket counter. Diagrams with top/front labelled.

2.5.3 Operations on Data Structures

LEARN IT

Whatever the shape, you keep meeting the same five operations:

Operation	Meaning	Quick picture
Insertion	Adding a new item into the structure.	PUSH on a stack; ENQUEUE on a queue; putting 40 into an array.
Deletion	Removing an item that is no longer needed.	POP; DEQUEUE; deleting a node from a list.
Traversal	Going through each item one by one.	Walking HEAD $\rightarrow \dots \rightarrow$ NULL; visiting every tree node.
Searching	Finding a specific item.	Linear search in an array; following links until 87 is found.
Sorting	Arranging items in a proper order (ascending or descending).	Bubble Sort on an array of marks.

WORKED EXAMPLE

Worked example: name the operation.

1. "Print every song from the start of the playlist." $\rightarrow$ **Traversal**.
2. "Add a print job at the rear." $\rightarrow$ **Insertion** (ENQUEUE).
3. "Undo the last typing action." $\rightarrow$ **Deletion** (POP).
4. "Is roll number 87 in the list?" $\rightarrow$ **Searching**.
5. "Arrange heights from shortest to tallest." $\rightarrow$ **Sorting**.

REMEMBER IT

Memorise **I-D-T-S-S**: Insert, Delete, Traverse, Search, Sort.

COMMON MISTAKE

Mistake: calling a PUSH "sorting" because the stack looks tidy.

Fix: PUSH is insertion. Sorting means putting values in ascending or descending order.

اردو میں سمجھیں

پر ڈیٹا سٹرکچر پر پانچ عمل ہوتے ہیں: داخل کرنا (Insertion)، حذف کرنا (Deletion)، ایک ایک کر کے گزرتا (Traversal)، تلاش (Searching) اور ترتیب (Sorting، صعودی یا نزولی)۔

CHECK YOURSELF

1. List the five operations.
2. Which operation is ENQUEUE?
3. Traversal vs searching — one difference?

Answers: 1. Insertion, deletion, traversal, searching, sorting. 2. Insertion (at the rear of a queue). 3. Traversal visits every item; searching stops when the target is found (or the structure ends).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Common DS operations include:

- a. Insert, delete, search, traverse
- b. Only compile
- c. Only ping
- d. Only print the logo

MCQ 2. Traverse means:

- a. Visit elements in some order
- b. Delete all
- c. Encrypt all
- d. Sort the OS

MCQ 3. Search cost depends on:

- a. The structure (array vs sorted vs hash vs tree)
- b. The PDF theme
- c. The teacher name
- d. The sprint

MCQ 4. Insert in a full array may:

- a. Fail or require resizing
- b. Always **$O(1)$** no matter what
- c. Create a graph edge automatically
- d. Pop a stack

MCQ 5. Delete from a queue happens at the:

- a. Front
- b. Rear only always
- c. Random index
- d. Root of a tree

MCQ 6. You choose a DS by the:

- a. Operations you will do most
- b. First letter of the course
- c. Colour
- d. NAND count

Extra short questions

1. List five operations.
2. Give costs qualitatively: array index vs list search.
3. What does traverse a tree mean?
4. Why "update" might be search+insert.

Extra long question

1. For array, stack, queue, tree: name the natural insert/delete ends and one costly operation. Table.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Create, insert, delete, search, traverse, update, sort (as applicable).
2. Array index **$O(1)$** ; unsorted list search **$O(n)$** .
3. Visit every node (pre/in/post or level order).
4. You must find the item first.

Long

1. Array: insert middle costly. Stack: top only cheap. Queue: rear in front out. Tree: search/insert depend on balance. 8-row table.

2.6 Identify data structures in problem scenarios

LEARN IT

To pick a structure, first see how the data is organised. If each item has a clear "before and after", it is **linear**. If it branches or connects to many items, it is **non-linear**.

2.6.1 Linear scenarios

- Storing student grades for a class → Array.** You usually know how many students there are, and all grades are the same type. Indices give rapid lookup.
- Music playlist / browser history → Linked list.** Every song knows what comes next. When one finishes, follow the pointer. Frequent insert/delete is cheap.
- Undo or Redo → Stack (LIFO).** Each typing or formatting action is PUSHed. Undo POPs the most recent action.
- Printer task scheduling → Queue (FIFO).** Documents print in the exact order they were received. First-come, first-served.

2.6.2 Non-linear scenarios

- Organisational chart of a college → Tree.** Principal is the root, then deans, then HODs, then teachers. Clear parent-child links.
- GPS navigation and maps → Graph.** Cities are nodes, roads are edges. No strict top-down hierarchy. Several routes can reach the same destination.

If the data...	Then it is...	Recommended structure
Follows "next" or "previous" logic	Linear	Array, Linked list
Follows "parent" and "child" logic	Non-linear	Tree
Follows "first-come, first-served" logic	Linear	Queue
Follows "connections / relationships" logic	Non-linear	Graph
Needs to be accessed in the reverse order of entry	Linear	Stack

WORKED EXAMPLE

Worked example: match five school problems.

Scenario	Structure	Why (one line)
Store 40 practical marks you already counted	Array	Fixed size, same type, need A[i] lookup.
A playlist where the teacher inserts a nasheed between two songs	Linked list	Insert by moving two pointers, no big shift.
Ctrl+Z after formatting a heading	Stack	Last action must come off first.
Students lining up for the photocopier	Queue	First in line is served first.
Friend links in a class WhatsApp "everyone knows everyone"	Graph	Any student can connect to any other.

REMEMBER IT

Decision order: reverse-of-entry? → stack. First-come? → queue. Parent-child? → tree. Many-to-many routes? → graph. Known list with indexes? → array. Lots of insert/delete in a sequence? → linked list.

COMMON MISTAKE

Mistake: choosing an array for Undo because "actions are stored in a list".

Fix: the **access rule** matters more than "it is a list". Undo is reverse order of entry → stack.

اُردو میں سمجھیں

اگر ہر عنصر کا واضح اگلا/پچھلا ہے تو ڈھانچہ لکیری ہے، اگر شاخیں یا کئی روابط ہیں تو غیر لکیری۔ نمبر: ایروء؛ پلے لسٹ/ایسٹری؛ لنکڈ لسٹ؛ انڈو: سٹیک؛ پرنٹن: کیو؛ کالج کا چارٹ: ٹری؛ GPS: گراف۔ فیصلے کی جدول یاد کر لیں۔

CHECK YOURSELF

- Which structure stores a class's grades, and why?
- Printer jobs: stack or queue? Why?
- "Needs reverse order of entry" maps to which structure?

Answers: 1. Array — known size, same type, fast index lookup. 2. Queue, because first-come, first-served. 3. Stack.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. An IDE is:

- One app to write, run and debug code
- A printer
- A sort
- A cable

MCQ 2. VS Code is:

- A popular editor/IDE
- A DBMS
- An OSI layer
- A gate

MCQ 3. The terminal/run button is for:

- Executing the program
- Drawing pie charts only
- Sending TCP SYN only
- K-maps

MCQ 4. Syntax highlighting helps you:

- See keywords and mistakes
- Sort faster
- Encrypt disks
- Route packets

MCQ 5. A file explorer in VS Code shows:

- Project files
- OSI layer 1
- NAND groups
- Beachhead users

MCQ 6. Debugging in an IDE uses:

- Breakpoints and variable watch
- Only bubble sort
- Only pie charts
- Only Agile stickers

Extra short questions

- Define IDE.
- Name four VS Code areas.
- Why not use Notepad only?
- What is an extension?

Extra long question

- Describe how you would create, run and debug a first Python file in VS Code. Include two interface elements from the notes.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

- Integrated Development Environment: editor + run + debug + files in one place.
- Activity bar/sidebar (files), editor, terminal, status bar (and extensions).
- No easy run, no highlighting, no debugger.
- An add-on (Python plugin) that teaches VS Code a language.

Long

- New file. .py, type print, Run Python, read terminal. Mention top menu, explorer, terminal panel. Breakpoint on a wrong variable.

2.7 Trace data retrieval in Array, List and Queue

LEARN IT
Tracing data retrieval means following the step-by-step path of how an algorithm finds or accesses data. Array, linked list and queue are all linear, but the **effort** to find an item is very different because of how they are organised inside.

2.7.1 Retrieval in an array

Every element has a fixed index from 0 to n-1 pointing at a spot in memory. If the algorithm already knows the index, it does not search: it **jumps** (direct access, **O(1)**).

Array = [10, 50, 20, 70, 90, 30]. Direct access: Array[2] = 20.

Loop iteration	Index	Array access	Value
1	0	Array[0]	10
2	1	Array[1]	50
3	2	Array[2]	20

```
Array = [10, 50, 20, 70, 90, 30]
print("Direct Array[2] =", Array[2])
for i in range(3):
    print(i + 1, ":", f"Array[{i}]", Array[i])
```

```
OUTPUT
Direct Array[2] = 20
1 8 Array[0] 10
2 1 Array[1] 50
3 2 Array[2] 20
```

2.7.2 Retrieval in a linked list

Linked lists have **no index**. The only way is to start at HEAD and follow links one node at a time (sequential access, **O(n)**).

List: [12] → [98] → [23] → [87] → [56] → NULL. Search key: 87.

Step	Node data	Compare with 87	Action
1	12	12 != 87	Move next
2	98	98 != 87	Move next
3	23	23 != 87	Move next
4	87	87 == 87	FOUND — output True

```
nodes = [12, 98, 23, 87, 56]
key = 87
found = False
for i, value in enumerate(nodes, start=0):
    print(f"Step {i}: Check Node {i} ({value}).")
    if value == key:
        found = True
        print("FOUND! Output: True")
        break
    print("Not equal. Move Next.")
print("Result:", found)
```

```
OUTPUT
Step 1: Check Node 1 (12).
Not equal. Move Next.
Step 2: Check Node 2 (98).
Not equal. Move Next.
Step 3: Check Node 3 (23).
Not equal. Move Next.
Step 4: Check Node 4 (87).
FOUND! Output: True
Result: True
```

Key 87 sits at the fourth node, so you paid four checks. An array that already knew index 3 would have paid one jump.

2.7.3 Retrieval in a queue

A queue is FIFO. You **cannot** retrieve from the middle or the rear. The oldest item at the Front is always DEQUEUED first.



You cannot pull B until A has left.

Queue A, B, C. ENQUEUE D joins the rear. DEQUEUE must remove A, not C or B.

```
from collections import deque
q = deque(["A", "B", "C"])
q.append("D")
print("After ENQUEUE D:", list(q))
print("DEQUEUE:", q.popleft())
print("now:", list(q))
```

```
OUTPUT
After ENQUEUE D: ['A', 'B', 'C', 'D']
DEQUEUE: A
now: ['B', 'C', 'D']
```

WORKED EXAMPLE

Worked example: three retrievals compared for "get 20 / get 87 / get A".

Structure	Target	Path	Can you jump?
Array [10, 50, 20, ...]	20 at index 2	Array[2] in one step, or scan 10, 50, 20	Yes, if the index is known
Linked list 12→98→23→87→...	87	Must visit 12, 98, 23, then 87	No
Queue A-B-C	A (the only legal retrieval)	DEQUEUE Front	No — never the middle

REMEMBER IT

Array jumps, list walks, queue waits. If the exam asks "why is array retrieval different?", write **direct access by index** versus **sequential links** versus **Front only**.

COMMON MISTAKE

Mistake: DEQUEUEing C because "C is the interesting letter", or reading linked-list node 4 as `list[3]` without walking.

Fix: show every hop in the table. Marks are for the path, not only for FOUND.

آرڈو میں سمجھیں

آرڈو میں اگر انڈیکس معلوم ہو تو سیدھی چھلانگ لگی ہے، جیسے `Array[2] = 20`۔ لنکڈ لسٹ میں HEAD سے ایک ایک نوڈ دیکھنا پڑتا ہے، 87 پونہی نوڈ پر ملا۔ کیوں سے صرف Front لنکڈ ہے، درمیان سے نہیں۔ A، B، C میں DEQUEUE یعنی A ہے۔

CHECK YOURSELF

1. What is Array[2] in [10, 50, 20, 70, 90, 30]?

2. How many nodes are checked before 87 is found in 12→98→23→87→56?

3. In queue A, B, C, can you retrieve B without removing A?

Answers: 1. 20. 2. Four (12, 98, 23, 87). 3. No — a queue only DEQUEUES the Front, so A must leave first.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Tracing retrieval means: a. Following how an algorithm finds/returns an item b. Deleting the DS c. Drawing Figma d. Running pip	MCQ 2. Array retrieval by index is: a. Direct: <code>a[i]</code> b. Walk from head always c. Pop until empty always d. BFS always	MCQ 3. Linked-list retrieval of the k-th is: a. Walk k links b. O(1) index c. A binary cut d. A hash	MCQ 4. Queue retrieval of the next client is: a. Dequeue front (FIFO) b. Pop top (LIFO) c. Random node d. Tree root	MCQ 5. You cannot take the middle of a queue without: a. Breaking FIFO (or using another DS) b. It is the definition of queue c. A pie d. A sprint	MCQ 6. A trace should show: a. Pointers/indices before and after b. Only the career names c. Only colours d. Only OSI
---	--	--	--	---	--

Extra short questions

- Trace getting `a[2]` from array [10,20,30,40].
- Trace finding 30 in a list 10→20→30.
- Trace queue A-B-C serve one.
- Why tracing retrieval is in the syllabus.

Extra long question

1. Same three names stored in array, list, queue. Trace "get the second person" in each. Who is not allowed in a pure queue?

Extra answer key

MCQs: 1. A. 2. A. 3. A. 4. A. 5. A. 6. A.

Short

- Index 2 → 30 in one step.
- Start 10 ←, next 20 ←, next 30 ←. Three visits.
- Dequeue A; front is B. Remaining B-C.
- So you can justify the DS with actual steps, not slogans.

Long

- Array `a[1]`: list two hops; queue: you may only legally take the front — getting the second requires dequeue then the new front, which also removes the first. That is the point.

Computational Thinking & Algorithms

16 single-page lessons · 5 Golden · read, close, say aloud, then sit the exam

Chapter in one look

- **Correctness:** output matches expected results for valid inputs. Test with a trace table (manual dry run) or stepwise reasoning (logic verification).
- A **trace table** is micro-level: line, variables, I/O, conditions. Best for small loops. The soldier heights [1.91, 2.01, 1.87, 1.57, 1.48] print Total = 3.
- **Stepwise reasoning** is macro-level: valid start, all cases, correct conditions, loop stays correct. Sum 1..N starts at 0; N = 0 stays 0; N = 5 becomes 15.
- **Clarity** (read, understand, modify) often beats a tiny speed gain. It has two parts: modularity and readability.
- **Modularity:** SRP (one job), loose coupling, abstraction (call Sort without knowing Bubble vs Selection). Split CalculateSum and FindAverage.
- **Readability:** names (`velocity = distance / time`), self-documentation, simple flow, consistent style.
- **Efficiency** = time + space. Drivers: number of steps (Big O), conditions, repetitions.
- **Big O:** $O(1)$ constant (`arr[0]`), $O(\log n)$ binary search, $O(n)$ linear search, $O(n \log n)$ Merge Sort, $O(n^2)$ nested loops / Bubble Sort. Quadratic fails as n grows.
- **Framework:** Does it scale? What is the bottleneck (deepest loop)? Is the extra complexity worth it? For $n = 50$, simple $O(n)$ can beat fragile $O(\log n)$.
- **Refinements:** meaningful names, modules, indentation, why-comments; shrink Bubble Sort's inner loop; better algorithms; cache constants; pick the right structure.
- **Data structure:** systematic organising of data. Linear (one after another): array, linked list, stack, queue. Non-linear: tree, graph.
- **Array:** contiguous, index from 0, direct access $O(1)$, shifting on mid-insert. **Linked list:** nodes of data+link, HEAD to NULL, sequential access.
- **Stack LIFO:** PUSH/POP at TOP (Undo). **Queue FIFO:** ENQUEUE rear, DEQUEUE front (printer).
- **Tree:** one root, parent-child (college chart). **Graph:** vertices + edges, no strict hierarchy (GPS, friends).
- Five operations: insert, delete, traverse, search, sort. Choose by next/previous, parent/child, FIFO, relationships, or reverse-of-entry.
- **Retrieval:** array can jump to an index; a list walks from HEAD; a queue only removes the Front.

Key terms

Term	Meaning	اردو
Correctness	The algorithm's output matches the expected results for valid inputs.	درستگی
Trace table	A paper dry-run table of steps, variables, I/O and conditions.	ٹریس ٹیبل
Dry run	Executing an algorithm by hand, line by line.	دستی اجرا
Stepwise reasoning	Checking each logic block's state; also called logic verification.	مرحلہ وار استدلال
Clarity	How easy an algorithm is to read, understand and modify.	وضاحت
Modularity	Breaking work into independent functions, classes or modules.	ماڈیولریتی
Single Responsibility Principle	Each module should do one specific task.	واحد ذمہ داری کا اصول
Coupling	How tightly modules depend on each other; loose coupling is the goal.	ماڈیولز کی جڑت
Abstraction	Hiding implementation details so a caller need not know the inner method.	تجربید
Readability	How easily another person understands the logic without running it.	پڑھائی پن
Efficiency	How well an algorithm uses time (speed) and space (memory).	کارکردگی
Big O notation	A description of how running time or memory grows as n grows.	بگ او نوٹیشن

Computational Thinking & Algorithms

75 marks · 1 hour 15 minutes · no notes · no phones

CHAPTER EXAM RULES

Instructions. Sit this exam after Lesson 16 and the recap. **Section A** — all 15 MCQs (1 mark). **Section B** — all 10 short questions (3 marks). **Section C** — attempt any **three** long questions (10 marks). At least one Section C answer must be a ★ Golden topic. Write in your notebook. Do not open the key until time is up.

Marks. $15 + 30 + 30 = 75$. A 5-mark / 10-mark answer needs Define · Explain · Example · Diagram · Working.

سبق اور خلاصے کے بعد یہ امتحان بغیر نوٹس کے حل کریں۔ وقت پورا ہونے پر کلید کھولیں۔

Section A — Multiple choice (15 × 1 = 15)

1. An algorithm is correct when:
 - a. it compiles with no syntax error
 - b. its output matches the expected results for the given input
 - c. it uses Big O notation
 - d. it has no comments
2. A trace table is best described as:
 - a. a paper execution that tracks variables line by line
 - b. a diagram of a tree
 - c. a type of graph edge
 - d. an $O(n^2)$ nested loop
3. Stepwise reasoning mainly checks:
 - a. printer ink levels
 - b. the colour of variable names
 - c. only the last printed line
 - d. the overall state of logic blocks
4. Which method is impractical if a loop runs 1,000 times?
 - a. Abstraction
 - b. A full trace table
 - c. Stepwise reasoning
 - d. Using a function
5. The Single Responsibility Principle says each module should:
 - a. focus on one specific task
 - b. print and save and calculate everything
 - c. avoid meaningful names
 - d. use nested IF only
6. Which assignment is more readable?
 - a. `v = d / t`
 - b. `x1 = a2 / b2`
 - c. `velocity = distance / time`
 - d. `a = b / c`
7. Accessing `arr[0]` is:
 - a. $O(n^2)$
 - b. $O(n \log n)$
 - c. $O(n)$
 - d. $O(1)$
8. Binary search on a sorted list is:
 - a. $O(\log n)$
 - b. $O(n^2)$
 - c. $O(1)$ always for any n
 - d. $O(n)$ because a loop exists
9. Bubble Sort and nested “compare every pair” loops are:
 - a. $O(1)$
 - b. $O(\log n)$
 - c. $O(n)$
 - d. $O(n^2)$
10. If n grows 10 times, a quadratic algorithm’s time grows about:
 - a. 2 times
 - b. 10 times
 - c. 100 times
 - d. log 10 times
11. LIFO is the rule of a:
 - a. queue
 - b. graph of cities
 - c. tree root only
 - d. stack
12. ENQUEUE adds an item at the:
 - a. Front of a queue
 - b. TOP of a stack only
 - c. Rear of a queue
 - d. root of a tree
13. A college organogram with a Principal at the top is modelled as a:
 - a. tree
 - b. queue
 - c. stack
 - d. contiguous array of roads
14. GPS maps (cities and roads, many possible routes) need a:
 - a. stack of undo actions
 - b. graph
 - c. strict one-parent tree only
 - d. single array index
15. To retrieve 87 from HEAD→12→98→23→87→56 you must:
 - a. jump with Array[87]
 - b. DEQUEUE the rear
 - c. follow four nodes from HEAD until 87 matches
 - d. POP from the bottom of a stack

Section B — Short questions ($10 \times 3 = 30$)

1. Define correctness of an algorithm. Name the two analysis methods used in this chapter.
2. List the components of a trace table. Why is a dry run useful for debugging?
3. Give four differences between a trace table and stepwise reasoning.
4. What is modularity? Explain SRP, coupling and abstraction in one sentence each.
5. State four ways to evaluate readability, with one example of a bad name and a good name.
6. Define Big O notation. Give one example each of $O(1)$, $O(\log n)$, $O(n)$, $O(n \log n)$ and $O(n^2)$.
7. Write the three questions of the efficiency evaluation framework and answer them for searching 50 unsorted names.
8. List four refinements for clarity and four for efficiency.
9. Define data structure. Distinguish linear from non-linear with examples.
10. Define stack and queue. Include LIFO/FIFO, PUSH/POP, ENQUEUE/DEQUEUE, and one real-life example of each.

Section C — Long questions (attempt any $3 \times 10 = 30$)

1. The soldier-eligibility algorithm uses AllHeights = [1.91, 2.01, 1.87, 1.57, 1.48] and counts heights from 1.57 to 1.91 inclusive. (a) Draw a complete trace table. (b) State the printed Total. (c) Explain in stepwise-reasoning style why the boundaries 1.57 and 1.91 are counted. (d) Write equivalent Python and its output.
2. Copy the monoblock average algorithm for A = [1, 2, 3, 4, 5] and rewrite it with CalculateSum and FindAverage. (a) Show that SRP, loose coupling and abstraction are improved. (b) Dry-run both versions to the same average. (c) Give two readability improvements you would still make to the functions.
3. Using the growth table idea, (a) fill n = 8 and n = 32 for $O(1)$, $O(\log_2 n)$, $O(n)$, $O(n \log_2 n)$ and $O(n^2)$. (b) Explain what happens if n becomes 10 times larger for $O(n)$ and for $O(n^2)$. (c) A library has 1,000,000 sorted IDs. Choose linear or binary search using the Scale / Bottleneck / Worth-it framework. (d) Show a 4-by-4 nested loop counts 16 operations in Python.
4. Draw a labelled stack and a labelled queue. Then trace: (a) PUSH Data1, PUSH Data2, PUSH Data3, POP, POP. (b) ENQUEUE 55, 76, 90, D then DEQUEUE. Include Python that matches both traces. (c) Why is Undo a stack and a printer a queue?
5. A college needs structures for: (i) 200 integer marks, (ii) a playlist with frequent insertions, (iii) Undo in the fee-office editor, (iv) photocopier jobs, (v) the Principal–Dean–HOD chart, (vi) roads between Sindh cities. Identify each structure using the decision guide, draw the tree and the graph, and then trace retrieval of 20 from Array = [10, 50, 20, 70, 90, 30] and of 87 from the linked list 12→98→23→87→56→NULL.

Answer key. Finish the Chapter Exam first, then mark. Do not peek while the clock is running.

پہلے امتحان مکمل کریں، پھر اس کلید سے نمبر لگائیں۔

Check yourself — answers

1. **2.1 Analyze algorithms for correctness.** Answers: 1. When its output matches the expected results for a given input (and does so for all valid inputs). 2. Trace tables and stepwise reasoning. 3. Because a logical error can still produce output — the wrong output.
2. **2.1.1 – 2.1.3 Trace Tables vs Stepwise Reasoning.** Answers: 1. Line/step number, variables, input/output, condition check. 2. 1.57 is the lower boundary (inclusive, so True); 1.48 is below 1.57 (False). 3. 3 is printed; Step 6 runs three times (for 1.91, 1.87 and 1.57).
3. **2.2 Evaluating the Clarity of an Algorithm.** Answers: 1. The algorithm is easy to read, understand and modify. 2. Modularity and readability. 3. Because a clear algorithm is easier to maintain, debug and extend over time.
4. **2.2.1 – 2.2.2 Modularity and Readability.** Answers: 1. One specific task. 2. CalculateSum (FindAverage only divides; the main program only prints). 3. All logic is in one block, so you cannot call “just the sum” from another algorithm.
5. **2.3 Assess Algorithm Efficiency.** Answers: 1. Time (speed) and space (memory). 2. Number of steps (Big O), number of conditions, number of repetitions.
6. **2.3.1 Number of Steps (Big O Notation).** Answers: 1. $O(1)$ arr[0]; $O(\log n)$ binary search; $O(n)$ linear search; $O(n^2)$ Bubble Sort / nested loops. 2. 1,024. 3. It makes about $\log n$ split-levels and does about n work on each level, so $n \times \log n$.
7. **2.3.2 Number of Conditions.** Answers: 1. Each condition must be evaluated True/False by the CPU. 2. A brute-force approach (and it reduces readability and speed). 3. AND short-circuits: the first part is False, so the division is never evaluated.
8. **2.3.3 Number of Repetitions.** Answers: 1. Poor — $O(n^2)$. 2. Each iteration halves the remaining problem, so the number of iterations is about $\log n$. 3. 25.
9. **2.3.4 The Efficiency Evaluation Framework.** Answers: 1. About 100× slower (10^2). 2. The most time-consuming section, usually the deepest loop. 3. Binary search is more complex to code and maintain, and for $n = 50$ the simplicity of $O(n)$ is practically better.
10. **2.4 Refinements to improve clarity and efficiency.** Answers: 1. average = totalSum / totalItems. 2. The already-sorted largest value at the end (shrink the inner loop). 3. Once, outside the loop, then reuse C inside.
11. **2.5 Concept of Data Structure.** Answers: 1. A systematic way of organising, storing and managing data for efficient access, modification and processing. 2. Linear: array, linked list, stack, queue. Non-linear: tree, graph. 3. It has exactly one predecessor and one successor.
12. **2.5.1 Linear Data Structures (Array, Linked List, Stack, Queue).** Answers: 1. The items sit in one continuous block, next to each other. 2. 12. 3. Later elements must all be shifted right to make room.
13. **2.5.2 Non-Linear Data Structures (Tree, Graph).** Answers: 1. The root. 2. One-to-many parent-child. 3. A family tree, and folders inside folders (or a college organogram).
14. **2.5.3 Operations on Data Structures.** Answers: 1. Insertion, deletion, traversal, searching, sorting. 2. Insertion (at the rear of a queue). 3. Traversal visits every item; searching stops when the target is found (or the structure ends).
15. **2.6 Identify data structures in problem scenarios.** Answers: 1. Array — known size, same type, fast index lookup. 2. Queue, because first-come, first-served. 3. Stack.
16. **2.7 Trace data retrieval in Array, List and Queue.** Answers: 1. 20. 2. Four (12, 98, 23, 87). 3. No — a queue only DEQUEUES the Front, so A must leave first.

Answer key

Multiple choice

1. b
2. a
3. d
4. b
5. a
6. c
7. d
8. a
9. d
10. c
11. d
12. c
13. a
14. b
15. c

Short questions — model answers

1. An algorithm is correct if its output matches the expected results for a given input (and does so for all valid inputs). The two methods are trace tables and stepwise reasoning (logic verification).
2. Components: line/step number, a column per variable, input/output, and a condition-check column. A dry run acts as the computer, so you see logical errors where the program runs but the values go wrong, and you can validate that the intention is met.
3. Trace table: small blocks/complex loops/beginners; high effort (bad for 1,000 iterations); catches arithmetic/line mistakes; micro-level exact values. Stepwise reasoning: large systems and proofs; moderate effort; catches conceptual/structural flaws; macro-level overall state.
4. Modularity is breaking an algorithm into smaller independent parts (functions, classes, modules). SRP: each module does one task. Coupling: modules should be loosely coupled so a change in one does not break others. Abstraction: hide inner details (call Sort without knowing Bubble vs Selection).
5. Naming conventions, self-documentation, simple control flow, consistency of indentation/spacing/casing. Bad: $v = d / t$. Good: `velocity = distance / time`.
6. Big O describes how time or memory grows as n grows. $O(1)$: `arr[0]`; $O(\log n)$: binary search; $O(n)$: linear search; $O(n \log n)$: Merge Sort; $O(n^2)$: Bubble Sort / nested loops.
7. Does it scale? $n = 50$ is tiny, even $O(n)$ is fine. Bottleneck? A single scan of 50 names. Worth it? Binary search needs sorted data and more complex code, so linear search is practically better.
8. Clarity: meaningful names; modules; structure/indentation; short why-comments. Efficiency: reduce extra steps (shrink Bubble Sort's inner loop); better algorithms (binary vs linear on sorted millions); avoid re-computing (cache $C = A+B$ outside the loop); appropriate data structures (array for direct access, tree for hierarchy).
9. A data structure is a systematic way of organising, storing and managing data for efficient access, modification and processing. Linear: sequence with one predecessor and one successor (array, linked list, stack, queue). Non-linear: hierarchical or network links (tree, graph).
10. Stack: LIFO; PUSH adds at TOP; POP removes from TOP; example Undo. Queue: FIFO; ENQUEUE adds at Rear; DEQUEUE removes from Front; example printer jobs.

Long questions — model answers

1. (a) Columns: Step, Total, Height, AllHeights, condition, output. Step 2 sets Total = 0. Step 3 stores the five heights. Then five loop cycles:

- 1.91 → True → Total 1
- 2.01 → False → Total stays 1
- 1.87 → True → Total 2
- 1.57 → True (boundary) → Total 3
- 1.48 → False → Total stays 3

(b) Printed Total = **3**.

(c) The condition is height ≥ 1.57 AND height ≤ 1.91 , so the closed interval includes both ends. 1.57 and 1.91 satisfy the inequalities; 2.01 and 1.48 do not. Initial Total = 0 is valid; each True branch adds one; False skips the add; after five items Total is the count of eligible soldiers.

(d) Python:

```
AllHeights = [1.91, 2.01, 1.87, 1.57, 1.48]
Total = 0
for height in AllHeights:
    if height >= 1.57 and height <= 1.91:
        Total = Total + 1
print(Total)
```

OUTPUT
3

2. (a) Monoblock: one sequence does the loop, the division and the print, so you cannot reuse the sum. Modular version: CalculateSum only adds; FindAverage only divides; main only prints. SRP holds. CalculateSum can be called from another program (loose coupling). Main does not see the loop (abstraction).

(b) Both compute total = 15 and avg = $15/5 = 3.0$.

```
def calculate_sum(A, n):
    total = 0
    for i in range(n):
        total = total + A[i]
    return total

def find_average(A, n):
    return calculate_sum(A, n) / n

print(find_average([1, 2, 3, 4, 5], 5))
```

OUTPUT
3.0

(c) Keep snake_case names; avoid 1-based I in Python; a one-line comment on why you divide by n (empty-list guard could be added later) is enough — do not comment every addition.

3. (a) Growth ($\log_2$):

n	$O(1)$	$O(\log n)$	$O(n)$	$O(n \log n)$	$O(n^2)$
8	1	3	8	24	64
32	1	5	32	160	1,024

(b) $O(n)$: 10× input → about 10× time. $O(n^2)$: 10× input → about 100× time, which fails for large sets.

(c) Scale: a million IDs; linear worst case is a million checks, binary about 20. Bottleneck: the search. Worth it: the IDs are already sorted and binary search is standard, so choose **binary search** $O(\log n)$.

(d)

```
n = 4
count = 0
for i in range(n):
    for j in range(n):
        count = count + 1
print(count)
```

OUTPUT
16

4. Draw the vertical stack with TOP on Data3 after three PUSHes, and the horizontal queue with FRONT on 55 and REAR on 90 (then D).

(a) Stack: [] → [Data1] → [Data1, Data2] → [Data1, Data2, Data3] → POP Data3 → POP Data2 → [Data1].

```
stack = []
stack.append("Data1")
stack.append("Data2")
stack.append("Data3")
print(stack.pop())
print(stack.pop())
print(stack)
```

OUTPUT
Data3
Data2
['Data1']

(b) Queue: [55] → [55, 76] → [55, 76, 90] → [55, 76, 90, D] → DEQUEUE 55 → [76, 90, D].

```
from collections import deque
q = deque()
q.append(55)
q.append(76)
q.append(90)
q.append("D")
print(q.popleft())
print(list(q))
```

OUTPUT
55
[76, 90, 'D']

(c) Undo must cancel the most recent action (LIFO → stack). A printer must respect arrival order (FIFO → queue).

5. (i) Array — known count, same type, index lookup. (ii) Linked list — next-pointer, cheap insert. (iii) Stack — reverse order of entry. (iv) Queue — first-come, first-served. (v) Tree — parent-child from the Principal root (Administrative Head / Academic Dean, then Accounts, HR, HOD Math, HOD CS). (vi) Graph — cities as vertices, roads as edges, alternative routes.

Array retrieval: Array[2] = 20 in one direct access (or after visiting 10 then 50 then 20 if you scan).

List retrieval: Step1 12 ≠ 87, Step2 98 ≠ 87, Step3 23 ≠ 87, Step4 87 = 87 FOUND True.

Self-assessment: I can...

- ☐ prove correctness with a full trace table, including boundary values.
- ☐ judge modularity (SRP, coupling, abstraction) and readability (N-S-S-C).
- ☐ apply the Scale / Bottleneck / Worth-it framework, including the $n = 50$ warning.
- ☐ define a data structure and classify linear vs non-linear.
- ☐ draw a tree and a graph and pick a structure from a scenario using the decision guide.
- ☐ write stepwise reasoning (start, loop, finish, edges) and compare it with a trace table.
- ☐ state $O(1)$, $O(\log n)$, $O(n)$, $O(n \log n)$ and $O(n^2)$ with examples and a growth table.
- ☐ refine an algorithm for names, modules, fewer loop steps and cached values.
- ☐ draw and trace an array, linked list, stack (PUSH/POP) and queue (ENQUEUE/DEQUEUE).
- ☐ trace retrieval: array jump, list walk from HEAD, queue Front only.